COM3031 Probabilistic Machine Learning : Week 1

Term 2, 2025-2026

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In this week

Lectures

- Introduction to the course structure
- Introduction to probabilistic machine learning
- Introduction to Bayesian Inference

Workshop

- Introduction to Coursework Assessment

Part I : Introduction to the course structure

Module Delivery

- **Lectures:** Fridays 08:35-9:25 and 10:35-11:25
- **Workshops:** Thursdays 13:35-14:25
- Office hours: Fridays 09:30-10:30 (by appointment, first floor, Innovation Centre I or Zoom)
- <u>No lectures and workshops in Week 11 (ten teaching weeks).</u>
- **Programming language**: Jupyter notebooks for demonstration.

Module Contents

- **Probabilistic modelling and inference**: Bayesian perspective; conjugate families; linear regression, generalized linear models; logistic regression, generative and discriminative models
- **Approximation Inference**: Laplace approximation; variational approximation; stochastic approximation (e.g., Markov Chain Monte Carlo methods)
- **Information theory**: information, entropy; coding; learning from an information-theoretic point of view.
- **Neural Networks:** Autoencoders and variational AEs, Bayesian Neural Networks
- **Learning in spatially and temporally connected models**: Hidden Markov models
- **Reinforcement learning**: Multi-armed bandits; finite Markov decision processes; on and off-policy learning.

Resources

- Module material - http://vle.exeter.ac.uk
- Bishop, C. ***Pattern Recognition and Machine Learning***, edition 1, Springer, 2006.
- Russell, S. and Norvig, P. ***Artificial Intelligence: A Modern Approach***, edition 4 Pearson, 2016.
- Mackay, D.J.C. ***Information Theory, Inference, and Learning***, edition 1, Cambridge, 2006.
- Murphy, K. ***Machine Learning: A Probabilistic Perspective***, MIT Press, 2012.
- Bayesian neural networks: A tutorial, Wesley Maddox, NYU.

Check the full reading list on the module page on ELE.

Workshops

Jupyter notebooks with practical examples

- Download the .ipynb file from ELE
- Open it on your local machine
- Or upload it to your Google Drive and open it with Google Colab
  - Colab Tutorial
  - Machine learning and Data Science examples with Colab
  - PyTorch tutorials (Webpage)

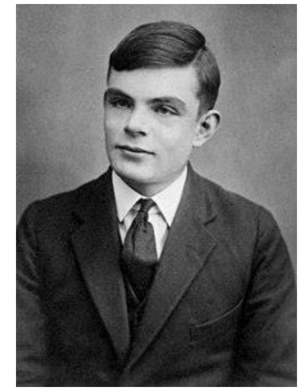
Assessment

- Formative assessment
  - Programming exercises covered during the workshops
  - Solutions will be provided
- Summative assessment
  - Coursework (30%) Assessment:
    - Available on ELE
    - Deadline: Feb 24, 2026
  - Written exam (70%) Assessment: 2 hours in May.

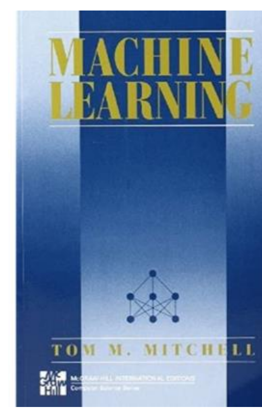
Any questions about the module structure?

Part II: Introduction to probabilistic machine learning

Machine Learning: a Quick Review



''Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's? If this were then subjected to an appropriate course of education one would obtain the adult brain.'' -Alan Turing (1950)



"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E." -Tom Mitchell (1997)

Machine Learning: a Quick Review

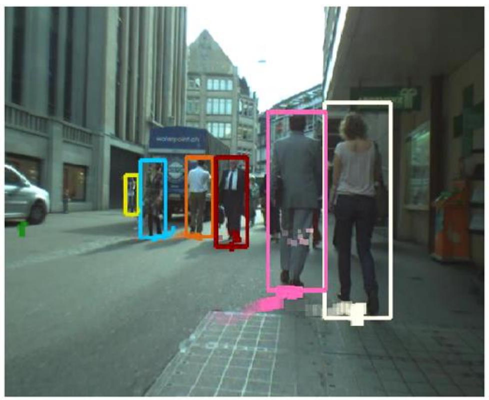
Machine Learning Paradigms

- **“Pure” Reinforcement Learning (cherry)**
  - The machine predicts a scalar reward given once in a while.
  - **A few bits for some samples**
- **Supervised Learning (icing)**
  - The machine predicts a category or a few numbers for each input
  - Predicting human-supplied data
  - **10→10,000 bits per sample**
- **Self-Supervised Learning (cake génoise)**
  - The machine predicts any part of its input for any observed part.
  - Predicts future frames in videos
  - **Millions of bits per sample**

(Source: Yann LeCun)

Machine Learning: a Quick Review

Human detection and tracking



Face detection and recognition

ChatGPT

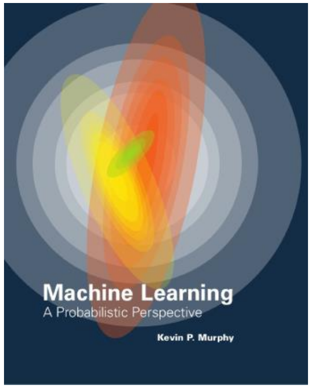
AlphaGo

Gemini

Image credits to MOTChallenge, [insightface](), Open AI and Google.

Why Probabilistic ML

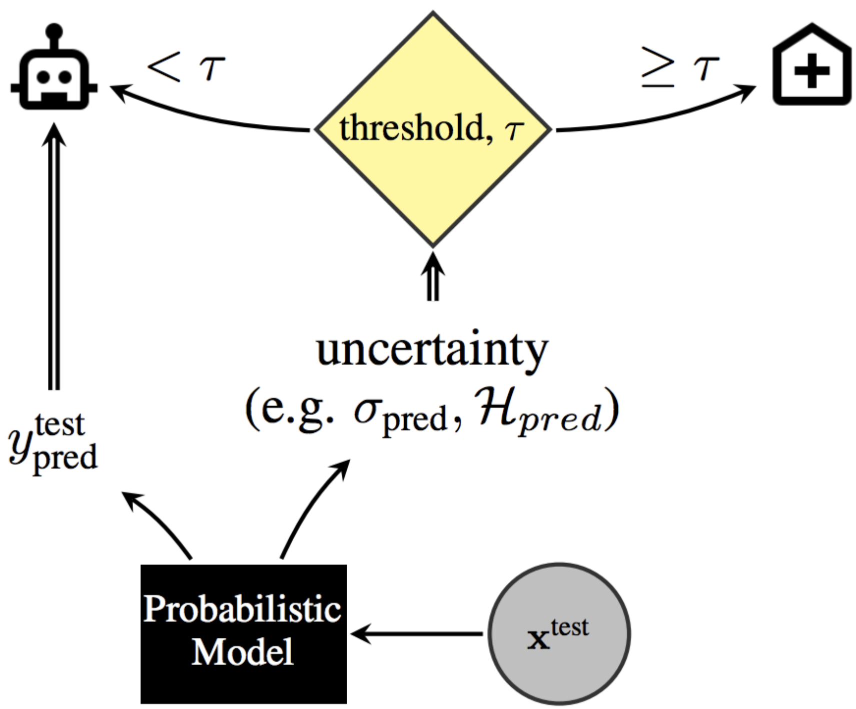
"The goal of machine learning is to develop methods that can automatically detect patterns in data, and then to use the uncovered patterns to predict future data, or to perform other kinds of ***decision marking under uncertainty****" –Kevin Murphy (2012)*



- Real-world involves uncertainty which needs to be quantified for efficient decision-making.
- R number of Covid - 19: number of cases directly generated by one person. If $R > 1$, the number of infections will increase.
- Decision making based on R number and involves some uncertainty.

Uncertainty in Neural Networks

Automated diagnosis: human-in-the-loop

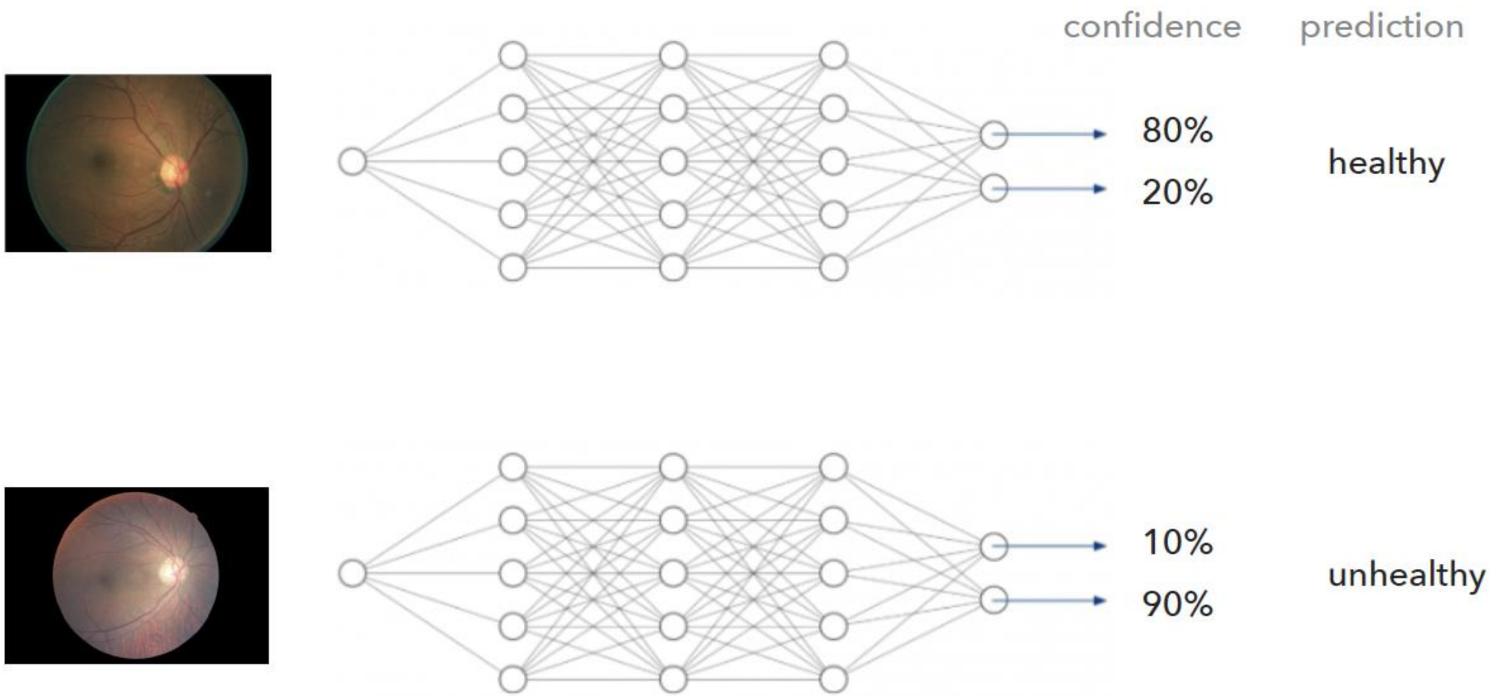


Source: Benchmarking Bayesian Deep Learning with Diabetic Retinopathy Diagnosis" by Angelos Filos et al.

Uncertainty in Neural Networks

University of Exeter

- Calibration



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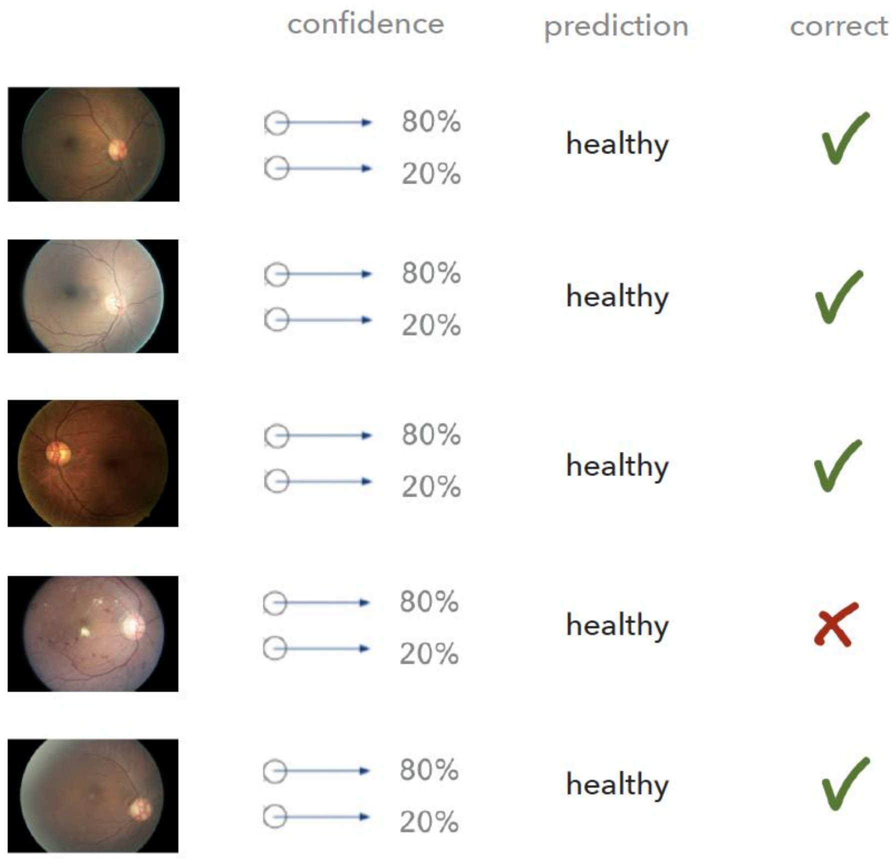
Uncertainty in Neural Networks

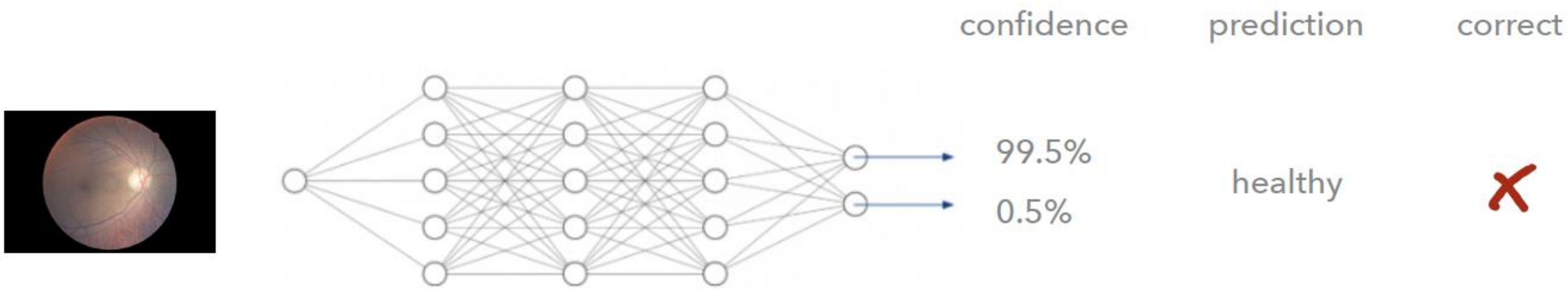
- Calibration



Source: Benchmarking Bayesian Deep Learning with Diabetic Retinopathy Diagnosis" by Angelos Filos et al.

Uncertainty in Neural Networks

- Neural networks are often over-confident in their predictions



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Two Types of Uncertainty

Aleatoric Uncertainty (Irreducible)

- Arises from **inherent randomness** in the data-generating process
- **Example (coin flip):**
  - Probability of heads or tails is 0.5
  - Flipping the coin more times does **not** reduce this uncertainty
- Cannot be eliminated, only **modelled probabilistically**

Epistemic Uncertainty (Reducible)

- Arises from **lack of knowledge** about the system
- **Example (coin already flipped):**
  - The outcome is fixed
  - You are uncertain only because you do not know the result
- Can be reduced with **more information or data**

Probabilistic Modelling

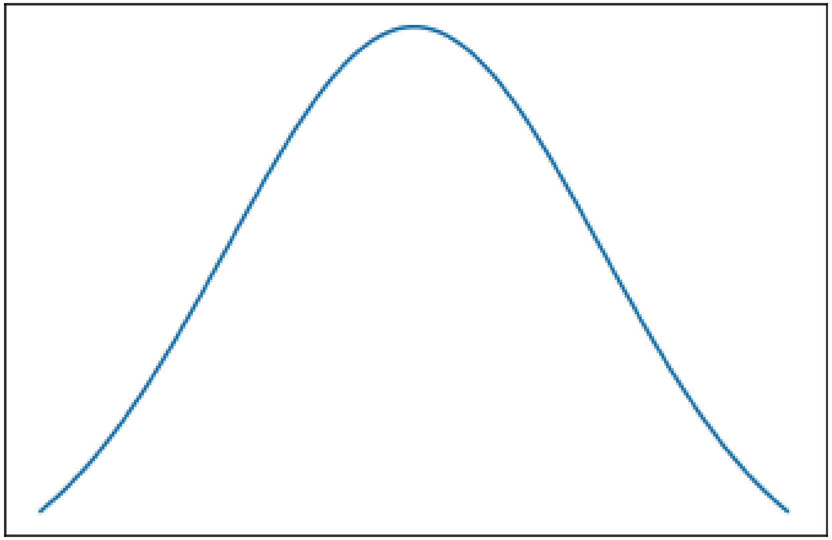
"A statistical model is a probability distribution constructed to enable inferences to be drawn or decisions made from data."

-A.C. Davison (2008)

- Model: Given parameters $\boldsymbol{\theta}$, the observations $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)$ are generated from a probability distribution $p(\boldsymbol{y}|\boldsymbol{\theta})$.
  - For a fixed $\boldsymbol{\theta}$, the model assigns probabilities (discrete) or probability densities (continuous) to possible outcomes $\boldsymbol{y}$.
- Goal: choose a suitable form for $p(\boldsymbol{y}|\boldsymbol{\theta})$ (e.g., Gaussian, Bernoulli, Poisson) that matches the data type and assumptions.

How do we choose a good form for $p(\boldsymbol{y}|\boldsymbol{\theta})$?

Probabilistic Modelling



Can you guess the distribution of the curve?

- Hint: It is known by different names and one of them is bell curve

Commonly used distributions

- Commonly used distributions and their pdfs in 1-D:

Distribution	Probability density function
Normal	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\frac{(y-\mu)^2}{\sigma^2}\right)$
Gamma	$\frac{\beta^\alpha y^{\alpha-1} e^{-\beta y}}{\tau(\alpha)}$
Poisson	$\frac{\lambda^y e^{-\lambda}}{y!}$
Binomial	$\binom{n}{y} p^y (1-p)^{n-y}$
Bernoulli	$p^y (1-p)^{1-y}$

Distribution	Typical use	Support
Normal	Linear-response data	real:$(-\infty, +\infty)$
Gamma	Exponential-response data	real:$(0, +\infty)$
Poisson	count data	integer: 0,1,2,. . .
Binomial	yes/no (or 1/0) data in n occurrences	integer: 0,1,. . . ,N
Bernoulli	yes/no (or 1/0) in 1 occurrence	integer: (0,1)

Probabilistic Modelling

- A common modelling choice is the Gaussian (normal) distribution.
- Model assumption: $p(y_i|\boldsymbol{\theta}) \sim \mathcal{N}(\mu, \sigma^2)$, $\boldsymbol{\theta} = (\mu, \sigma)$
  - A univariate normal (or Gaussian) distribution has two parameters: mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$.
- Single observation model (for fixed $\boldsymbol{\theta}$):

$$p(y_i|\boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mu)^2}{2\sigma^2}\right)$$

  - This describes how likely different values of $\boldsymbol{y_i}$ are given fixed parameters $\boldsymbol{\theta}$.

Likelihood

- **Dataset:** Given parameters $\boldsymbol{\theta}$, assume $(y_1, y_2 \ldots, y_n)$ are independent and identically distributed (IID), this allows us to write the joint data distribution as, $p(\boldsymbol{y}|\boldsymbol{\theta}) = \prod_{i=1}^{n} p(y_i|\theta)$
- **Likelihood function**: once the data $(y_1, y_2 \ldots, y_n)$ are observed:

$$L(\boldsymbol{\theta}) = p(\boldsymbol{y}|\boldsymbol{\theta}) = \prod_{i=1}^{n} p(y_i|\theta)$$

- Likelihood depends on the model - two different models for the same data $\boldsymbol{y}$ will typically have two different likelihoods
- **Inference:** use the likelihood to infer the unknown parameters.

Probabilistic Inference

Likelihood inference:

- Treat parameters $\boldsymbol{\theta}$ as fixed but unknown
- used the likelihood $p(\boldsymbol{y}|\boldsymbol{\theta})$ to infer $\boldsymbol{\theta}$
- find parameter values that best explain the observed data

$$\hat{\boldsymbol{\theta}} = \underset{\theta}{\operatorname{argmax}}\, p(\boldsymbol{y}|\boldsymbol{\theta})$$

Bayesian inference:

- Specify likelihood $p(\boldsymbol{y}|\boldsymbol{\theta})$
- Select a distribution for $p(\boldsymbol{\theta})$
- Combine them using Bayes' rule:
- **Goal**: infer a **posterior distribution** over parameters

$$p(\theta|\boldsymbol{y}) \propto p(\boldsymbol{y}|\theta)p(\theta)$$

Maximum Likelihood Estimation

- Find the model parameters $\boldsymbol{\theta}$ that best explain the observed data
- Maximise the likelihood to find the model parameters i.e.

$$\widehat{\boldsymbol{\theta}} = \underset{\theta}{\operatorname{argmax}}\, L(\boldsymbol{\theta}) \text{ or } \widehat{\boldsymbol{\theta}} = \underset{\theta}{\operatorname{argmax}} \prod_{i=1}^{n} p(y_i|\theta)$$

- argmax of $L(\theta)$ is the same as the argmax of $\log L(\theta)$ - makes the maths simpler:

$$\log L(\boldsymbol{\theta}) = \log \prod_{i=1}^{n} p(y_i|\theta) = \sum_{i=1}^{n} \log p(y_i|\theta)$$

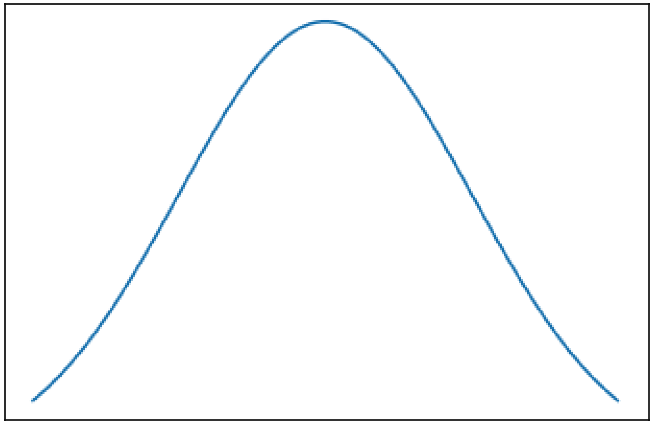
$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} \;\; \log L(\theta) = \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^{n} \log p(y_i|\theta)$$

Maximum Likelihood Estimation

- Let us assume the data is normally distributed and $\boldsymbol{\sigma}$ is known i.e.

$$p(y_i|\boldsymbol{\theta}) \sim \mathcal{N}(\mu, \sigma^2) \text{ for } \boldsymbol{i} = 1, \ldots, n$$

- Maximise the likelihood to get the parameter $\boldsymbol{\theta} = \boldsymbol{\mu}$



(Demonstration – on board)

Maximum Likelihood Estimation

Another example

- Flip a coin $\boldsymbol{n}$ times
- If heads: $\boldsymbol{y} = \mathbf{1}$; else $\boldsymbol{y} = \mathbf{0}$
- Number of heads: $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)$

Find the parameter $\boldsymbol{\theta}$ i.e. probability of getting heads

- Note: $p(y = 0) = 1 - p(y = 1)$

(Demonstration – on board)

Part III: Introduction to Bayesian Inference

Bayesian Inference

- **Likelihood: $p(\boldsymbol{y}|\theta)$:** the probability (or probability density) of the observed data $\boldsymbol{y} = (y_1, \ldots, y_n)$, given model parameters $\boldsymbol{\theta}$
- **Prior: $p(\theta)$:** represents uncertainty or beliefs about the parameter ☒ before observing any data.
- **Posterior: $p(\theta|\boldsymbol{y})$:** updated probability distribution over parameters after observing the data which can quantify the uncertainty in the parameters $\boldsymbol{\theta}$
- Predictive distribution: For a new observation $\boldsymbol{y}'$, predictions $\boldsymbol{p}(\boldsymbol{y}'|\boldsymbol{y})$ are obtained by averaging over parameter uncertainty and therefore provides **uncertainty-aware predictions.**

Prior

Prior distribution:

- Represents our beliefs or knowledge about model parameters before observing any data
- Can encode domain knowledge or reasonable constraints
- Can rule out implausible values of parameters
- Influences the posterior, especially when data are limited

How to choose a prior

- **Rule out impossible values:** Use the prior to encode basic constraints (e.g. probabilities must be between 0 and 1).
- **Non-informative** – Normal distribution with a constant mean and very large variance - $\boldsymbol{p(\theta)}$ is approximately same for all values of $\boldsymbol{\theta}$
- **Weakly informative** – Normal distribution with a constant mean and a large variance (e.g. 50-100) - $\boldsymbol{p(\theta)}$ same for $\boldsymbol{\theta}$ which are plausible
- **Informative** – Normal distribution with a constant mean and very small variance

Prior

- Let us assume the data $\boldsymbol{y} = (y_1, \ldots, y_n)$ is normally distributed (let as assume $\boldsymbol{\sigma}$ is given and $\boldsymbol{\theta} = \boldsymbol{\mu}$) i.e.

$$p(y_i|\mu) \sim \mathcal{N}(\mu, \sigma^2) \text{ for } \boldsymbol{i} = \mathbf{1}, \ldots, \boldsymbol{n}$$

- We also assume that the parameter $\boldsymbol{\mu}$ also follows a normal distribution - Prior (or belief before looking at the data) i.e.

$$p(\mu) \sim \mathcal{N}(\mu_0, \sigma_0^2)$$

- How to find the $\boldsymbol{p}(\boldsymbol{\mu}|\boldsymbol{y})$?

Bayesian Inference

- Bayes' rule:

$$p(\theta|\boldsymbol{y}) = \frac{p(\boldsymbol{y}|\theta)p(\theta)}{p(\boldsymbol{y})}$$

- The marginal distribution of y is for the normalisation and can also be written as:

$$p(\boldsymbol{y}) = \int p(\boldsymbol{y}|\theta)p(\theta)d\theta$$

- Because $\boldsymbol{p(y)}$ is constant with respect to $\boldsymbol{\theta}$

$$p(\theta|\boldsymbol{y}) \propto p(\boldsymbol{y}|\theta)p(\theta)$$

Maximum A Posteriori Estimation

Bayesian posterior:

$$p(\theta|\boldsymbol{y}) \propto p(\boldsymbol{y}|\theta)p(\theta)$$

Taking logs (for easier optimisation):

$$\log p(\theta|\boldsymbol{y}) \propto \log p(\boldsymbol{y}|\theta) + \log p(\theta)$$

- Instead of Maximum likelihood estimation (MLE), we now do the Maximum A Posteriori Estimation (MAP) to get the most likely parameter values
- $\hat{\theta}_{MAP} = \underset{\theta}{\operatorname{argmax}} \left[\log p(\theta|\boldsymbol{y}) \right] = \underset{\theta}{\operatorname{argmax}} \left[\log p(\boldsymbol{y}|\theta) + \log p(\theta)\right]$

Bayesian Inference

Recall the posterior,

$$p(\theta|\boldsymbol{y}) = \frac{p(\boldsymbol{y}|\theta)p(\theta)}{p(\boldsymbol{y})} \propto p(\boldsymbol{y}|\theta)p(\theta)$$

MAP: $\hat{\theta}_{MAP} = \underset{\theta}{\text{argmax}}\, p(\theta|\boldsymbol{y}) == \underset{\theta}{\text{argmax}}\, p(\boldsymbol{y}|\theta)p(\theta)$

(beacuase $\boldsymbol{p(y)}$ does not depend on $\boldsymbol{\theta}$)

However, the product of likelihood and prior: $\boldsymbol{p(y|\theta)p(\theta)}$ cannot be used to make probabilistic statements and is not a normalised probability density

Bayesian Inference

Demonstration on board for normal distribution

Conjugate Priors

Conjugate priors: A prior is ***conjugate*** to a likelihood if the posterior has the **same functional form** as the prior.

Parameter	Prior Distribution
Bernoulli	Beta
Binomial	Beta
Poisson	Gamma
Normal (fixed variance)	Normal for the mean
Normal (fixed mean)	Gamma prior for the inverse variance
Exponential	Gamma
Multinomial	Dirichlet

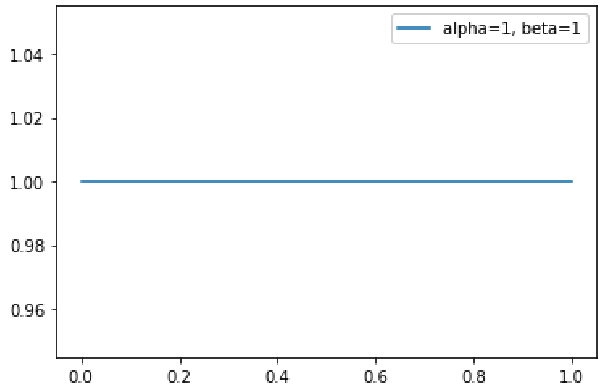
- Gives a **closed-form posterior**
- Avoids expensive numerical integration
- Makes Bayesian inference **fast and tractable**

Bayesian Inference

Another example

- $y = 0$: tails and $y = 1$ heads
- $\theta \in [0, 1]$
- What is the prior on θ
- Beta distribution: $p(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}$

Prior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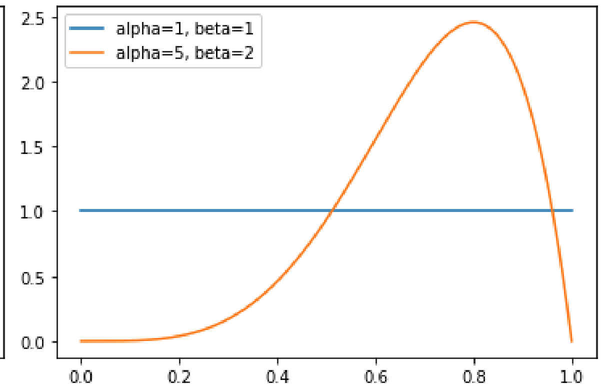
1.04
1.02
1.00
0.98
0.96
0.0 0.2 0.4 0.6 0.8 1.0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2.5 2.0 1.5 1.0 0.5 0.0
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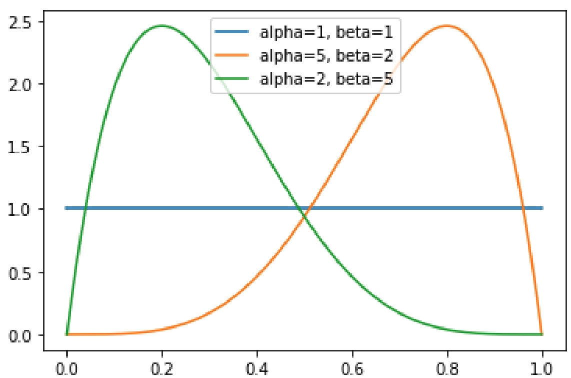
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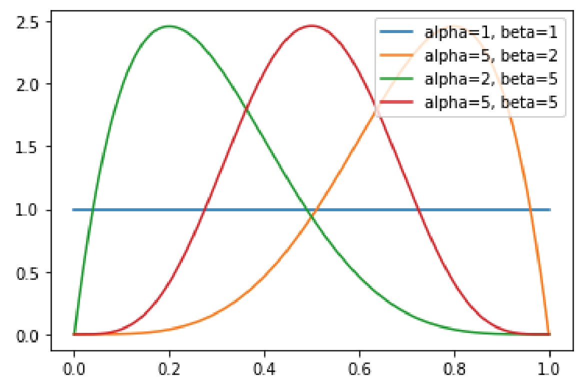
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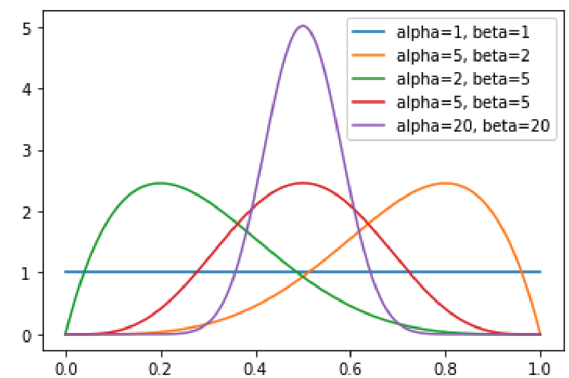
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alpha=5, beta=2
alpha=2, beta=5
alpha=5, beta=5
alpha=20, beta=20
5 4 3 2 1 0
0.0 0.2 0.4 0.6 0.8 1.0

Maximum A Posteriori Estimation

Demonstration on board – Posterior for binomial-beta distribution

Bayesian Inference

Advantages of **posterior distribution**:

- **Uncertainty-aware prediction (posterior predictive)**: We can predict a new observation y′ via complete posterior predictive distribution i.e. $p(y'|\boldsymbol{y})$, which accounts for both data uncertainty (aleatoric), and parameter uncertainty (epistemic).
- **Credible intervals for parameters**: We can compute the probability that a parameter lies in a range.
- **Model checking** (posterior predictive checks): We can simulate new datasets from the fitted model and compare them to the real data.

Bayesian Inference

Point estimates of $\boldsymbol{\theta}$:

- MAP:

$$\theta_{\text{MAP}} = \underset{\theta}{\operatorname{argmax}}\, p(\theta|\boldsymbol{y})$$

Conjugate priors:

- e.g., beta-binomial model
- Posterior available in close form

Approximation (when exact inference is unavailable):

- Laplace approximation
- Variational methods
- Stochastic approximation, e.g. Monte Carlo Markov Chain methods